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Course: CSE

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Course Design

Name of the Academic Program: M.Tech. in CSE

Course Code: CS3.304 Title of the Course: Advanced Operating Systems

L-T-P: 3-1-1. Credits: 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge

Digital Logic Design, Computer Organization, Knowledge of a programming language, Introduction to operating systems

2. Course Outcomes (COs)

Objective: The operating system abstracts the machine part of the given computer system in terms of simple services (system calls) by hiding the details of the machine (hardware). Traditionally, operating system concepts were developed for single processors (machines). Currently, shared memory, shared disk, and shared-nothing architectures (computer networks) are employed to build massive cloud computing environments by exploiting multiple processor cores and commodity machines to manage huge diverse computing loads, such as search engines, weather forecasting, and e-commerce services. The objectives of this course are to study the key concepts that were evolved for building modern operating systems.

The course Outcomes (COs) are as follows.

After completion of this course successfully, the students will be able to,

CO-1. Explain the concepts of several modern computer operating systems (SOLARIS, LINUX, WINDOWS, MAC, Android,...)

CO-2: Implement the task on the top of given operating system, efficiently, based on process and thread framework.

CO-3. Prescribe the appropriate scheduling/synchronization/memory management/virtual memory/protection module for a given application.

CO-4. Architect the new application by selecting the appropriate system calls of the given operating system services.

CO-5. Develop a distributed application on the multi-processor machine or multiple commodity processors connected through a network.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2
CO2	3	3	3	2	2	-	1	-	2	2	2	3	2	1	2	2

CO3	3	3	3	2	2	2	1	-	3	2	2	3	3	2	1	2
CO4	2	2	3	2	2	3	2	-	3	2	2	2	3	3	2	3
CO5	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2

Note '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus

- Unit 1: Introduction, Computer System Hardware Review, Operating System Structures (9 hours)
- Unit 2: Process and thread management, CPU scheduling, Process Synchronization, Deadlocks (12 hours);
- Unit 3: Overview of Computer Networking (6 hours)
- Unit 4: Memory management, Virtual memory (6 hours);
- Unit 5: Basics of file systems, RAID and Storage Structures, Hadoop Distributed File System, Map Reduce, Cloud Computing: models and technologies, Virtual Machines. (10 hours)
- Five mini projects related to the above syllabus will be done by students in the laboratory

References:

1. Silberschatz, Operating System Concepts 10Ed Global Edition (Pb 2023)
2. Computer Networks, 6e by Tanenbaum (Author)
3. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things Kai Hwang, Geoffrey C. Fox, Jack J. Dongarra, Elsevier, 2012
4. Operating Systems, 9e by William Stallings (Author)
5. Modern Operating Systems, 5th Edition, Tanenbaum (Author)
6. Research Papers (Key research papers related to UNIX, Virtual machines, smart phone OSs, distributed OSs.)

5. Teaching-Learning Strategies in brief

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing 5 mini-projects in laboratory by the students

6.Assessment methods and weightages in brief

Two Class Room tests: 10 marks; Mid Semester Examination in theory: 20 marks , End Semester Examination in Theory: 40 marks, Assessment of 5 mini projects in Laboratory: 30 marks
